

1. Hardware Specifications & Port Density

The NetCore Nexus-9000 is a high-density, low-latency 1RU data center switch featuring 48x 100GbE QSFP28 ports and 8x 400GbE QSFP-DD uplink ports. It delivers a total non-blocking switching capacity of 6.4 Tbps and a packet forwarding rate of 4.7 Bpps.

Parameter	Specification
Switching Capacity	6.4 Tbps (Full Duplex)
Forwarding Rate	4.7 Billion Packets/sec
Port Configuration	48x 100G QSFP28 + 8x 400G QSFP-DD
Buffer Size	64 MB Shared Dynamic Buffer
Latency	450 nanoseconds (cut-through)

2. Power and Cooling Requirements

Maximum power consumption under full 100% traffic load is 450 Watts, with a typical idle power draw of 185 Watts. The chassis supports 1+1 redundant hot-swappable 80-PLUS Platinum power supplies and 5+1 redundant fan trays with front-to-back airflow.

3. Reliability Metrics

Mean Time Between Failures (MTBF) is rated at 350,000 operating hours at 25C ambient temperature. The system operates within an ambient temperature envelope of 0C to 40C.